

JKBOSE 2026

CLASS 11TH

Relations and Functions

MATHS

LECTURE - 06

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TOPICS TO BE COVERED

1

Exercise Discussion 2.3

2

Related Problems.

$$f(x) = \sqrt{x^2 + 5x + 6} \rightarrow (\text{Domain})$$

EXERCISE – 2.3

Question

Which of the following relations are functions? Give reasons. If it is a function, determine its domain and range.

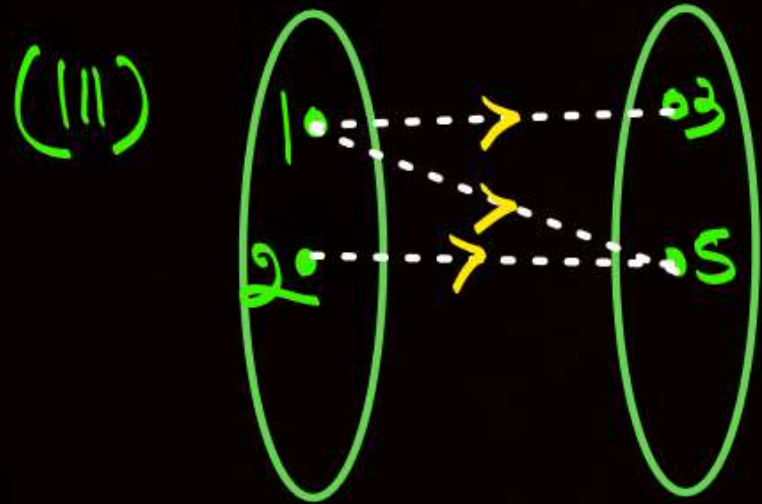
- (i) $\{(2, 1), (5, 1), (8, 1), (11, 1), (14, 1), (17, 1)\}$
- (ii) $\{(2, 1), (4, 2), (6, 3), (8, 4), (10, 5), (12, 6), (14, 7)\}$
- (iii) $\{(1, 3), (1, 5), (2, 5)\}$.

(Sol) (i)

It is a function with
 Domain = $\{2, 5, 8, 11, 14, 17\}$
 Range = $\{1\}$

(ii)

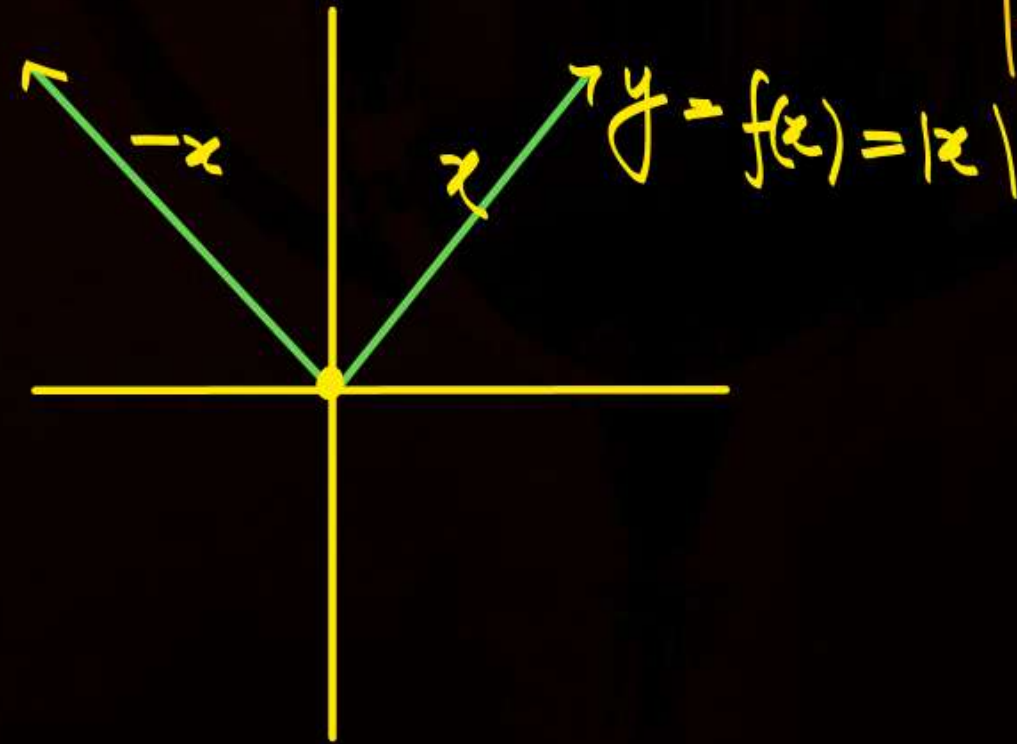
It is a function with
 Domain = $\{2, 4, 6, 8, 10, 12, 14\}$
 Range = $\{1, 2, 3, 4, 5, 6, 7\}$



It is not a function as the image of 1 is not unique.

$$* f(x) = |x| = \begin{cases} x & ; x \geq 0 \\ -x & , x < 0 \end{cases} \quad |2| = 2.$$

$$|-2| = -(-2) = 2$$



Question

Find the domain and range of the following real functions:

(i) $f(x) = -|x|$

(ii) $f(x) = \sqrt{9 - x^2}$

(i)(sol) Given function is:

$$f(x) = -|x| = \begin{cases} -x & ; x \geq 0 \\ -(-x) & ; x < 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -x & ; x \geq 0 \\ x & ; x < 0 \end{cases}$$

Since the function is defined for all values of $x \in \mathbb{R}$

, therefore Domain = $\mathbb{R}$.

For any real number x ,

$$|x| \geq 0$$

$$\Rightarrow -|x| \leq 0$$

$$\Rightarrow f(x) \leq 0$$

So Range = $(-\infty, 0]$

(ii)(sol) Given function is;

$$f(x) = \sqrt{9-x^2}$$

For $f(x)$ to be real, we have;

$$9-x^2 \geq 0$$

$$\Rightarrow 3^2 - x^2 \geq 0$$

$$\Rightarrow (3+x)(3-x) \geq 0$$

$$\Rightarrow -(x-3)(x+3) \geq 0$$

Multiplying on b/s by -1 , we get;

$$(x-3)(x+3) \leq 0$$

We know that the product of two numbers is negative if one of them is negative. So;

$$\text{Case I: } x-3 \leq 0 \text{ \& } x+3 \geq 0$$

$$\Rightarrow x \leq 3 \text{ \& } x \geq -3$$

$$\Rightarrow -3 \leq x \leq 3$$

$$\Rightarrow x \in [-3, 3]$$

$$\text{Case II: } x-3 \geq 0 \text{ \& } x+3 \leq 0$$

$$\Rightarrow x \geq 3 \text{ \& } x \leq -3$$

$$\Rightarrow \text{No such } x \text{ exists.}$$

$$\text{So } x \in [-3, 3] \cup \phi$$
$$\Rightarrow x \in [-3, 3]$$

$$\text{Therefore Domain} = [-3, 3]$$

* Wavy Curve method:

$$\begin{array}{c} \oplus \quad \ominus \quad \oplus \\ \hline -3 \quad 3 \end{array}$$

$$x \in [-3, 3]$$

$$\text{So, Domain} = [-3, 3]$$



Further, for $x \in [-3, 3]$, so

$$0 \leq x^2 \leq 9$$

$$\Rightarrow 0 \geq -x^2 \geq -9$$

$$\text{or } -9 \leq -x^2 \leq 0$$

$$\Rightarrow -\cancel{9} + \cancel{9} \leq 9 - x^2 \leq 0 + 9$$

$$\Rightarrow 0 \leq 9 - x^2 \leq 9$$

$$\Rightarrow \sqrt{0} \leq \sqrt{9 - x^2} \leq \sqrt{9}$$

$$\Rightarrow 0 \leq f(x) \leq 3$$

$$\Rightarrow f(x) \in [0, 3]$$

Therefore Range = $[0, 3]$

Hence the result.

Question: Find the Domain of $f(x) = \sqrt{x^2 + 5x + 6}$.

(Sol) Given function is

$$f(x) = \sqrt{x^2 + 5x + 6}$$

For $f(x)$ to be real, we have

$$x^2 + 5x + 6 \geq 0$$

$$\Rightarrow x^2 + 2x + 3x + 6 \geq 0$$

$$\Rightarrow x^2 + 2x + 3x + 6 \geq 0$$

$$\Rightarrow x(x+2) + 3(x+2) \geq 0$$

$$\Rightarrow (x+3)(x+2) \geq 0$$

We know that the product of two numbers is positive if both are positive or negative.

So, **Case 1st:** $x+3 \leq 0$ & $x+2 \leq 0$

$$\Rightarrow x \leq -3 \text{ & } x \leq -2$$

$$\Rightarrow x \leq -3$$

$$\Rightarrow x \in (-\infty, -3]$$



Case 2nd: $x+3 \geq 0$ & $x+2 \geq 0$

$$\Rightarrow x \geq -3 \text{ & } x \geq -2$$

$$\Rightarrow x \geq -2$$

$$\Rightarrow x \in [-2, \infty)$$



Therefore from both Cases, we get;

$$x \in (-\infty, -3] \cup [-2, \infty)$$

$$\text{So, Domain} = (-\infty, -3] \cup [-2, \infty)$$

Question

A function f is defined by $f(x) = 2x - 5$. Write down the values of

(i) $f(0)$

Homework.
#

(ii) $f(7)$

$$f(0) = 2 \times 0 - 5 = -5.$$

(iii) $f(-3)$.

Question

The function 't' which maps temperature in degree Celsius into temperature in degree

Fahrenheit is defined by $t(C) = \frac{9C}{5} + 32$.

(Conversion formula)

Find.

✓ (i) $t(0)$

✓ (ii) $t(28)$

✓ (iii) $t(-10)$

(iv) The value of C, when $t(C) = 212$

(i) (Sol) $t(0) = \frac{9 \times 0}{5} + 32$

$\Rightarrow t(0) = 32^\circ F$

$0^\circ C = 32^\circ F$

(ii) $t(28) = \frac{9 \times 28}{5} + 32 = \text{---} ??$

(iii) $t(-10) = \frac{9 \times -10}{5} + 32 = -18 + 32 = 14^\circ F$

(iv)(sol) Given $t(c) = 212$

$$\Rightarrow \frac{9c}{5} + 32 = 212.$$

$$\Rightarrow \frac{9c + 160}{5} = 212.$$

$$\Rightarrow 9c + 160 = 1060.$$

$$\Rightarrow 9c = 1060 - 160$$

$$\Rightarrow 9c = 900$$

$$\Rightarrow c = \frac{900}{9} = 100.$$

$$\Rightarrow \boxed{c = 100}$$

Question

Find the range of each of the following functions.

(i) $f(x) = 2 - 3x, x \in \mathbf{R}, x > 0.$

(ii) $f(x) = x^2 + 2, x$ is a real number.

(iii) $f(x) = x, x$ is a real number. (Range = $\mathbf{R}$)

(i) (Sol) For $x > 0$, we have

$$3x > 0$$

$$\Rightarrow -3x < 0$$

$$\Rightarrow 2 - 3x < 0 + 2$$

$$\Rightarrow \boxed{f(x) < 2}$$

So Range = $(-\infty, 2)$

(ii) (Sol) For any real number
 $x^2 \geq 0$

$$\Rightarrow x^2 + 2 \geq 0 + 2$$

$$\Rightarrow \boxed{f(x) \geq 2}$$

So, Range = $[2, \infty)$

Question

If $f(x) = ax + b$ where a and b are integers, $f(-1) = -5$ and $f(3) = 3$, then a and b are equal to

☒ 1 $a = -3, b = -1$

☒ 2 $a = 2, b = -3$

☒ 3 $a = 0, b = 2$

☒ 4 $a = 2, b = 3$

(Sol) Given function is;

$$f(x) = ax + b, a, b \in \mathbb{Z}.$$

Here $f(-1) = -5$ & $f(3) = 3$

$$\Rightarrow -a + b = -5 \quad \text{--- (1)}$$

$$\& 3a + b = 3 \quad \text{--- (2)}$$

Subtracting (1) & (2), we get

$$-a + b - 3a - b = -5 - 3$$

$$\Rightarrow -4a = -8$$

$$\Rightarrow a = \frac{-8}{-4} = 2.$$

$$\Rightarrow a = 2.$$

From (2), we get;

$$3 \times 2 + b = 3.$$

$$\Rightarrow 6 + b = 3$$

$$\Rightarrow b = 3 - 6$$

$$\Rightarrow b = -3.$$

Question



The domain of the function f defined by $f(x) = \frac{1}{\sqrt{x - |x|}}$ is

1 $\mathbb{R}$

2 $\mathbb{R}^+$

3 $\mathbb{R}^-$

4 None of these

Homework Problem:

$$x - |x| > 0$$

$$x > |x| \quad (\text{Homework})$$

★★★★★ (BAAZ) → Shaheen

Thank
You

↓
Udna
Higher & Higher